PROBLEM UNDERSTANDING

1. Formulate a technical project description. Explain your product (preferably as a flowchart)

The **Cosmic Navigators AI-Powered Space Debris Tracking System** is built to make space safer by detecting, tracking, and optimizing the movement of space debris. Instead of relying on real-time data, we use historical satellite imagery and archived data from space agencies like NASA and ESA. With the power of machine learning and computer vision, our system helps satellite operators and aerospace companies avoid collisions and plan safer missions.

What We Aim to Do:

1. Build smart AI models that can spot space debris using old satellite data.
2. Use machine learning to optimize debris path management.
3. Create an easy-to-use dashboard that shows debris movement and generates reports.
4. Keep data secure and follow international space regulations.

Flowchart: How Our Space Debris Detection System Works

1. **Collect Data**
   - We gather historical satellite images and debris information from NASA, ESA, and other agencies.
2. **Prepare the Data**
   - The data is cleaned, labeled, and normalized to make sure it's accurate and ready for analysis.
3. **Detect Debris**
   - Using computer vision techniques like Convolutional Neural Networks, we identify debris in satellite images.
4. **Classify Debris**
   - Detected debris is categorized based on its size, material, and how dangerous it is.
5. **Optimize Paths**
   - Machine learning algorithms optimize debris movement paths to minimize collision risks.
6. **Assess Risks**
   - We analyze the likelihood of collisions with satellites and recommend actions to avoid them.
7. **Visualize & Report**
   - The results are shown on an interactive dashboard, and detailed reports are generated for satellite operators.
8. **Improve Over Time**
   - Our AI models keep learning from new data to improve accuracy.

2. Build a customer journey map to visualize how customers interact with your product/ service/ system.

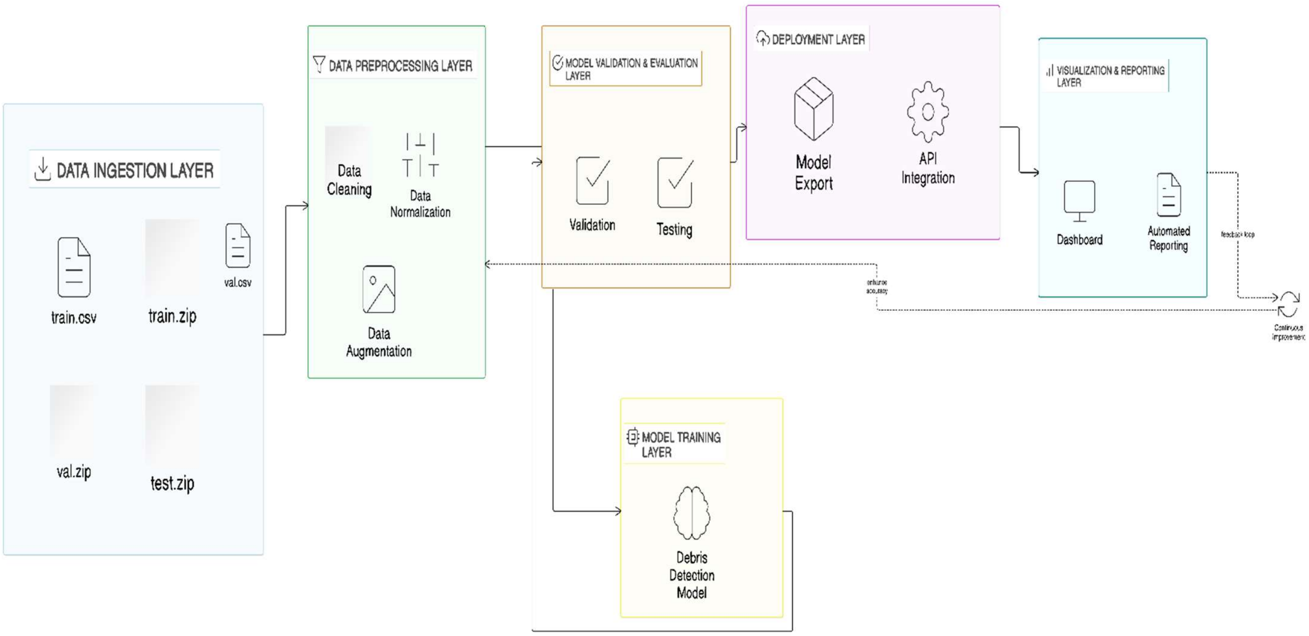
Customer Journey Map: Cosmic Navigators Space Debris Tracking System

1. **Awareness Stage**
   Customer Goal: Find an effective solution for tracking and managing space debris.
   Touchpoints: Discovering Cosmic Navigators through aerospace conferences, online articles, or partnerships with space agencies.

Pain Points: Many potential customers might not even know AI-based debris tracking exists or how it can help them.

2. **Decision Stage**
   Customer Goal: Choose the best debris tracking system that fits their needs.
   Touchpoints: Watching product demos, reviewing technical documents, and comparing with other solutions.
   Pain Points: The technology might seem complicated, and customers may worry about how it will fit with their existing systems.
3. **Onboarding Stage**
   Customer Goal: Get the system up and running smoothly.
   Touchpoints: Accessing integration guides, receiving hands-on technical support, and participating in training sessions.
   Pain Points: Setting up new technology can be tricky, and there's always a learning curve for teams getting used to new tools.
4. **Usage Stage**
   Customer Goal: Use the system to detect, classify, and manage debris effectively.
   Touchpoints: Regularly using the dashboard for real-time debris tracking, getting automated alerts, and analyzing reports.
   Pain Points: There might be occasional system slowdowns or delays, especially when handling large amounts of data.
5. **Analysis & Optimization Stage**
   Customer Goal: Continuously improve satellite paths to reduce the risk of collisions.
   Touchpoints: Using data visualization tools, running predictive analyses, and leveraging AI for path optimization.
   Pain Points: The data can be complex, and customers may need help making sense of it all to get the best results.
6. **Feedback & Improvement Stage**
   Customer Goal: Share feedback to help improve the system.
   Touchpoints: Submitting feedback through the dashboard, joining user feedback sessions, and benefiting from regular updates.
   Pain Points: Customers may want new features rolled out faster or wish for more customization to meet their unique needs.
7. **Retention & Advocacy Stage**
   Customer Goal: Keep using the system and recommend it to others in the industry.
   Touchpoints: Renewing service contracts, participating in case studies, and sharing testimonials.
   Pain Points: Ensuring the system stays reliable over time and keeps up with the latest technology developments.

3. Build a visualization of high-level system architecture.



4. Describe all the core features of your product

Core Features of the Product

1. **Seamless Data Integration**
   - We pull in historical satellite imagery and debris data from reputable sources like NASA and ESA, creating a solid foundation for our tracking and detection processes.
2. **Smart Debris Detection**
   - Our AI leverages Convolutional Neural Networks (CNNs) to accurately detect and identify space debris, outperforming traditional tracking methods.
3. **Comprehensive Debris Classification**
   - The system classifies debris by size, material, and risk level, providing precise insights into potential threats.
4. **Path Optimization for Safer Missions**
   - Machine learning algorithms help optimize debris trajectories, reducing the likelihood of collisions with operational satellites.
5. **Proactive Risk Assessment**
   - We predict potential collisions and provide actionable recommendations, helping satellite operators stay ahead of threats.
6. **User-Friendly Interactive Dashboard**
   - Our intuitive dashboard offers real-time debris monitoring, easy-to-understand visualizations, and comprehensive reporting tools.
7. **Continuous Learning and Improvement**
   - The AI continuously learns from new data, refining its detection, classification, and prediction capabilities over time.
8. **Flexible API Integration**
   - Our APIs allow seamless integration with existing satellite management tools, making it easy for space agencies and private operators to adopt the system.

9. **Global Compliance and Sustainability**
    - We ensure that all tracking and data management processes align with international space regulations and sustainability standards, supporting responsible space exploration.

5. Briefly define the project scope.

Project Scope

The **Project Scope** for our AI-powered space debris tracking system outlines the boundaries and goals of the project, ensuring everyone involved is on the same page. It defines what we'll deliver, what we won't, and how we'll get there, acting as our roadmap throughout development.

Key Elements of Project Scope:

- **Objectives:** To create an AI-driven system that accurately detects, classifies, and predicts space debris movement, helping satellite operators avoid collisions.
- **Deliverables:** The final product will include an AI-powered debris detection model, a user-friendly dashboard, automated reporting tools, and API integration for seamless use.
- **Milestones:** Major milestones include completing AI model training, launching the dashboard, conducting integration tests, and rolling out the final system.
- **Constraints:** We're working within limits like the availability of reliable historical data, budget considerations, and strict international space regulations.
- **Assumptions:** We're assuming that historical data from agencies like NASA and ESA will be accessible and that satellite operators will be ready to integrate our system.
- **Requirements:** The system needs to provide accurate debris tracking, reliable collision predictions, and an intuitive user interface.
- **Boundaries:** We won't be dealing with real-time data collection from new satellites or handling the physical removal of debris.
- **Acceptance Criteria:** Success means our system delivers high detection accuracy, clear risk assessments, and smooth integration with existing satellite management tools.

Why This Scope Matters:

- **Guidance:** It gives the team a clear path, making sure everyone knows what we're building and how we'll get there.
- **Resource Allocation:** Helps us use time, budget, and resources wisely without unnecessary detours.
- **Risk Management:** By defining what's in and out of scope, we spot risks early and avoid surprises down the road.
- **Stakeholder Alignment:** Keeps everyone—from developers to satellite operators—on the same page with clear expectations.
- **Control:** Acts as our safety net, preventing scope creep and ensuring the project stays focused and on track.

6. Briefly discuss about the dataset available that can be used in the project.

Dataset for the Project

To build and refine our AI models, we're working with a rich dataset specifically curated for space debris detection. This dataset gives our models the depth they need to recognize and predict debris with precision.

What’s in the Dataset?

1. train.zip - A collection of 20,000 satellite images showcasing space debris. Each image is labeled and tied to entries in train.csv.
2. train.csv - This file acts as a map for the training images, linking ImageID to corresponding debris locations through bboxes (bounding boxes).
3. val.zip - A smaller set of 2,000 images used to fine-tune and validate our model’s performance.
4. val.csv - Similar to the training file, it provides metadata to help validate model accuracy, including ImageID and bounding boxes.
5. test.zip - This set of 5,000 images is used to test how well the model performs on new, unseen data.